Complex Analysis Homework # 9

1. Show that any automorphism of $\mathbb{H}$ preserves the hyperbolic metric (that is, it's an *isometry* for the metric). Are these the only isometries?

We show that automorphisms of $\mathbb{H}$ are isometries by direct calculation. Let $f \in PSL_2(\mathbb{R})$, $z = x + iy \in \mathbb{H}$, and $w = f(z)$. If we allow $|dz|^2$ to denote $|dz|^2 = dx^2 + dy^2$ then our metric is $\frac{|dz|^2}{y^2}$. To show that f is an isometry we need to show that

$$\frac{|dw|^2}{Im(f)^2} = \frac{|dz|^2}{y^2}.$$

We have that $dw = f'(z)dz$ and so let us compute the derivative $f'(z)$. Recall that $f(z) = \frac{az+b}{cz+d}$ for $a, b, c, d \in \mathbb{R}$ and $ad - bc = 1$. Then using the quotient rule gives

$$\begin{aligned} f'(z) &= \frac{a}{cz+d} - \frac{c(dz+b)}{(cz+d)^2} \\ &= \frac{acz + ad - acz - bc}{(cz+d)^2} \\ &= \frac{1}{(cz+d)^2}. \end{aligned}$$

And so

$$\frac{|dw|^2}{|dz|^2} = \frac{1}{|cz+d|^2}.$$

On the other hand let us compute $Im(f)^2$. We have

$$\begin{aligned} w &= \frac{az+b}{cz+d} \\ &= \frac{(az+b)\overline{(cz+d)}}{|cz+d|^2} \\ &= \frac{(az+b)(c\bar{z}+d)}{|cz+d|^2} && a, b, c, d \in \mathbb{R} \\ &= \frac{ac\,|z|^2 + adz + bc\bar{z} + bd}{|cz+d|^2}. \end{aligned}$$

Now, notice that the denominator is a real number and that the only part of the numerator

which may contain imaginary terms is $adz + bc\bar{z}$. Consider

$$\begin{aligned} adz + bc\bar{z} &= ad(x+iy) + bc(x-iy) \\ Im(adz + bc\bar{z}) &= ady - bcy \\ &= y. \end{aligned}$$

That is,

$$\frac{Im(w)^2}{Im(z)^2} = \frac{1}{|cz+d|^2}\frac{y^2}{y^2} = \frac{1}{|cz+d|^2}.$$

We have thus shown that automorphisms of $\mathbb{H}$ are isometries of the hyperbolic metric.

Now let $f : \mathbb{H} \to \mathbb{H}$ be an isometry of the upper half plane with the hyperbolic metric. We have that, $ds^2 = \frac{dx^2+dy^2}{y^2}$ which is a conformal transform of the euclidean metric. Which means that its isometries must be conformal. We have seen that the only conformal functions of $\mathbb{C}$ are the holomorphic functions and the antiholomorphic functions. An antiholomorphic function cannot be a function $\mathbb{H} \to \mathbb{H}$. And so f is a holomorphic function of $\mathbb{H}$.

Now that we know f is holomorphic, the calculation above shows that f must satisfy

$$\frac{|f'(z)|^2}{Im(f(z))} = \frac{1}{Im(z)}.$$

If $f(x,y) = u(x,y) + iv(x,y)$ this expression gives us a condition on v.

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2. Show that any vertical line in $\mathbb{H}$ is a geodesic for the hyperbolic metric (i.e. for any pair of points p, q on the line, the shortest path between the points is along the line). Show that any semicircle in $\mathbb{H}$ that meets the boundary $\mathbb{R}$ at right angles is a geodesic.

Let p, q be two points of the upper half plane with the same real part. If γ is a curve in the upper half plane, the length of γ under the hyperbolic metric is given by contracting the tangent vectors against the metric along the curve. That is if $\gamma : [0, 1] \to \mathbb{H}$ with $\gamma(0) = p$ and $\gamma(1) = q$ then the hyperbolic length of this curve is

$$L = \int_a^b \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))} dt,$$

where $g_{(x,y)}(u, v)$ means the contraction of tangent vectors u, v in the tangent space at $p = (x, y) \in \mathbb{H}$. (I am using some conceptual understanding from physics here, is this also the way one would say this in metric geometry?)

If γ is not the vertical line connecting p, q then it moves a nonzero amount parallel to the real line. For a fixed imaginary value, the hyperbolic length of a horizontal line is given by a scaling of its euclidean length. That is, any distance travelled in the "real direction" by γ adds a positive value to the length of our curve. In particular this means that the geodesic connecting p, q is given by the curve which does not travel in the real direction at all — this is namely the vertical line connecting p, q.

Now suppose that p, q two points no longer lying on a vertical line. We saw in the previous question that any automorphism of $\mathbb{H}$ is a hyperbolic isometry. In particular, it must bring geodesics to geodesics. Then, it is sufficient to show that there exists an automorphism of $\mathbb{H}$ which brings p, q to a vertical line. For a given p, q there's a unique semicircle meeting the real axis at a right angle specified by three points in the complex plane p, q, x where $x \in \mathbb{R}$. By triple transitivity of the Mobius transformations, there's then a unique Mobius transformation M mapping $(x, p, q) \mapsto (0, \tilde{p}, \tilde{q})$ where $\tilde{p}, \tilde{q} \in \mathbb{H}$ and are colinear.

A priori our transform $M \in PGL_2(\mathbb{C})$, however, if we can show that $M \in PSL_2(\mathbb{R})$ then are done. Since $M \in GL_2(\mathbb{C})$ it has determinant $D \neq 0$. We can scale M by any $\lambda \in \mathbb{C}$ to acquire an equivalent transformation. But $|\lambda M| = \lambda^2 D$, and so scaling M by $1/\sqrt{D}$

means that M can be considered to have determinant 1 without loss of generality.

Recall that the automorphisms of $\mathbb{H}$ are a subgroup of $PSL_2(\mathbb{R}) \leq PGL_2(\mathbb{C})$ and so, we need to argue that the transformation found above has real entires.

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3. Find a metric on Δ that is invariant under any automorphism of Δ.

We found in problem 1 that the automorphisms of the upper half plane are exactly the automorphisms of the hyperbolic metric on the upper half plane. We also know a biholomorphism between $\mathbb{H}$ and Δ and so we can use this to compute a candidate metric on Δ which is preserved under any automorphism of Δ.

We saw in class a biholomorphism $\Delta \to \mathbb{H}$ via

$$h(z) = i \cdot \frac{1+z}{1-z}.$$

This map has holomorphic inverse $f : \mathbb{H} \to \Delta$ via

$$f(z) = \frac{z-i}{z+i}.$$

Now let us denote the hyperbolic metric on $\mathbb{H}$ by $ds^2 = (dx^2 + dy^2)/y^2 = |dz|^2/y^2$. Then using the relation $dw = f'(z)dz$ where $w = f(z)$ we can compute how the metric transforms under f. Using the quotient rule we get that

$$f'(z) = \frac{2i}{(z+i)^2}.$$

And so

$$|dz|^2 = \frac{|z+i|^4}{4} |dw|^2 .$$

Now if $z = x + iy \in \mathbb{H}$, let us compute $y = Im(z)$ in terms of w. We have that

$$\begin{aligned} x + iy &= i \cdot \frac{1+w}{1-w} \\ &= i \cdot \frac{1 - \bar{w} + w - |w|^2}{|1-w|^2} \\ y &= \frac{1 - |w|^2 + Re(w - \bar{w})}{|1-w|^2} \\ y &= \frac{1 - |w|^2}{|1-w|^2}. \end{aligned}$$

Note that if $a \in \mathbb{C}$ then $a - \bar{a}$ is a purely imaginary number.

Now we can compute a candidate metric

$$\begin{aligned} ds^2 &= \frac{1}{y^2}\left|dz\right|^2 \\ &= \left(\frac{\left|1-w\right|^2}{1-\left|w\right|^2}\right)^2 \frac{\left|z+i\right|^4}{4}\left|dw\right|^2. \end{aligned}$$

Now notice that ssince $w = \frac{z-i}{z+i}$ we have that $1 - w = \frac{2i}{z+i}$. And so $\left|z+i\right|^2 = 4/\left|1-w\right|^2$. Now, we are left with

$$ds^2 = \frac{1}{\left(1-\left|w\right|^2\right)^2}\left|dw\right|^2,$$

where $w \in \Delta$. This is our candidate metric on Δ.

Recall that the automorphisms of Δ are of the form

$$g(z) = e^{i\theta}\phi_a(z),$$

for some angle θ and where ϕ_a is a Blaschke map. If we show that rotations and all ϕ_a are isometries of our proposed metric then compositions of such maps (i.e. all automorphisms of Δ) will also be isometries. Notice that the conformal factor of our candidate metric depends only on $\left|w\right|^2$, the magnitude of our point in Δ. Thus, it is certainly preserved under rotations. The last thing to compute is whether the Blaschke maps all satisfy

$$\frac{\left|\phi_a'(z)\right|^2}{\left(1-\left|\phi_a(z)\right|^2\right)^2} = \frac{1}{\left(1-\left|z\right|^2\right)^2},$$

for all $z \in \Delta$ and all $a \in \Delta$.

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4. Let $f : \Delta \to \Delta$ be holomorphic. Determine the possibilities for the set of fixed points of f.

Let f be as given in the question. First suppose that $f(0) = 0$. We show that if this function has a nonzero fixed point then it is the identity. Recall the Schwarz lemma for such a function states

$$|f(z)| \leq |z| \text{ for all } z \in \Delta \qquad \text{and} \qquad \left|f'(0)\right| \leq 1,$$

and if equality is achieved in either for some nonzero $z_0 \in \Delta$, then f is a rotation. Suppose f has a fixed point, i.e. $f(z_0) = z_0$ for some $z_0 \in \Delta$. But then we have $|f(z_0)| = |z_0|$ and so f is a rotation, $f(z) = e^{i\theta}z$. However, since $f(z_0) = z_0$ and $z_0 \neq 0$ it must be that $e^{i\theta} = 1$, i.e. f is the identity. That is, a nonidentity holomorphic f with $f(0) = 0$ only fixes the origin.

Now let f be general (i.e. drop the assumption $f(0) = 0$). I claim that if f has more than one fixed point then it must be the identity on Δ. Suppose that f has two distinct fixed points $f(a) = a$ and $f(b) = b$. Now let $g : \Delta \to \Delta$ be defined by

$$g = \phi_a \circ f \circ \phi_a^{-1},$$

where $\phi_a : \Delta \to \Delta$ is the Blaschke map (the automorphism of the disc) which exchanges 0 and a. Since $f(a) = a$ this function satisfies $g(0) = 0$. Notice that $\phi_a(b)$ is also a nonzero fixed point of g. And so by the argument above, g is the identity. Now conjugating g by ϕ_a implies that $f(z) = z$ is the identity (recall that $\phi_a^{-1} = \phi_a$). Thus, we have shown that if f has two distinct fixed points then it must be the identity on Δ. In other words, a non-identity holomorphic function on Δ has at most a single fixed point.

We show that there exists a holomorphic function on Δ whose set of fixed points is the empty set. Let $x \in \mathbb{R}$ be nonzero and consider the Blaschke map

$$\phi_x = \frac{z - x}{1 - xz}.$$

The fixed point equation $\phi_x(z) = z$ is equivalent to

$$0 = x(1 - z^2).$$

However the solutions to this equation in z are only for $z \in \partial\Delta$. That is, this equation has no solutions in Δ and so ϕ_x has no fixed points in Δ. Recall that ϕ_x is a holomorphic function on Δ since it is an automorphism of Δ.

We have shown that if f is non identity then it can only have a single fixed point. The constant map $f : \Delta \to \Delta$ via $f(z) = a$ for some $a \in \Delta$ has a fixed point at a. If $f : \Delta \to \Delta$ is the identity restricted to the disc has every point as a fixed point.

Overall, the possibilities for the set of fixed points are Δ, $\{a\}$ where $a \in \Delta$, and $\varnothing$.

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5. Let $U \subset \mathbb{C}$ be open and let $\mathcal{F}$ be a family of real differentiable functions $U \to \mathbb{C}$. Suppose there exists a real number M such that all four partials of f at every point are bounded in absolute value by M. Show $\mathcal{F}$ is equicontinuous. Does the reverse implication hold, i.e. does equicontinuity imply boundedness of partials?

We say that a family of functions $\mathcal{F}$ is equicontinuous if for all $\epsilon > 0$ there exists some $\delta = c(\epsilon)$ such that for all $f \in \mathcal{F}$ and for all $z, w \in U$ such that $|z - w| < \delta$ then $|f(z) - f(w)| < \epsilon$. That is, every f is uniformally continuous, and the single delta works for all functions in the family.

Suppose we have a condition on the partials of $f \in \mathcal{F}$ as given in the question. For $x \in U$, since U is open, we know that there is a ball containing x contained in U. Thus there's a line segment connecting x to sufficiently close by points and we have

$$f(y) - f(x) = \int Df(\gamma)(y - x)dt,$$

where γ is some line segment connecting x to y in the sufficiently small ball about x, and where Df is the Jacobian of f. Now we can use the estimate

$$|f(y) - f(x)| \leq \sup_{z \in \gamma} |Df(z)| \, |y - x| \, .$$

Here the norm on Df is the operator norm. The condition given on the question bounds the entries of the jacobian by M. This is enough to bound the operator norm by some other constant $\tilde{M}$. Now if we are given $\epsilon > 0$ we have shown that choosing

$$|y - x| < \frac{\epsilon}{\tilde{M}}$$

is sufficient to bound $|f(y) - f(x)| < \epsilon$ for all $x, y \in U$ and for all $f \in F$. Hence $\mathcal{F}$ is equicontinuous

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6. Show that $f(z) = -\frac{1}{2}(z + 1/z)$ gives a biholomorphism from the upper half disc to the upper half plane.

First, we need to verify that the proposed f has outputs to the upper half plane. And then, if we can construct an inverse function, then we have shown that the given function is a biholomorphism $\tilde{\Delta} \to \mathbb{H}$ where $\tilde{\Delta}$ denotes the upper half disc.

First we show that the proposed function outputs to $\mathbb{H}$. Let $z \in \Delta$, we need to show that $Im(f(z)) > 0$. Recall that $Im(f(z)) = (f(z) - \overline{(}f(z)))/2i$. A quick computation gives that

$$Im(f(z)) = \frac{1}{2i}\left(f(z) - \overline{f(z)}\right) = -\frac{1}{2}Im(z)\left(1 - \frac{1}{|z|^2}\right).$$

However, recall that $z \in \Delta$ and so, $1 - 1/|z|^2 < 0$. Thus, $Im(f(z)) > 0$ and the proposed function outputs to $\mathbb{H}$ for all $z \in \Delta$, and in particular, for all $z \in \tilde{\Delta}$.

Now, we show that the proposed function has an inverse $\mathbb{H} \to \Delta$. Let $w \in \mathbb{H}$, we want to show that there is a unique $z \in \tilde{\Delta}$ such that $w = -1/2(z + 1/z)$. Rearranging this equation gives

$$0 = z^2 + 2wz + 1.$$

As a quadratic equation, this has two roots $z_0, z_1 \in \mathbb{C}$. Recalling then that we can factor our quadratic as $0 = (z - z_0)(z - z_1)$, the coefficients from the above give the following relations on the roots

$$z_0 z_1 = 1 \tag{1}$$

$$z_0 + z_1 = -2w. \tag{2}$$

This first equation gives us that z_0 and z_1 are inverses of each other. I claim that neither z_0, z_1 lie on S^1. Suppose $z_0 \in S^1$ then equation 1 would imply that $z_1 \in S^1$ and that z_0, z_1 are conjugate to each other. But then their sum is real. And then equation 2 cannot be satisfied since $w \in \mathbb{H}$ and so $-2w$ has strictly negative imaginary part. Hence, neither z_0, z_1 lie on S^1.

Now, equation 1 implies that one of the roots lies in Δ and the other root lies in $\mathbb{C} \setminus \Delta$, since $z \mapsto 1/z$ exchanges the unit circle and its complement. Let's say that $z_0 \in \Delta$. Now

I claim that equation 2 can only be satisifed if in fact $z_0 \in \tilde{\Delta}$. If we write $z_0 = re^{i\theta}$, $r < 1$ then $z_1 = z_0^{-1} = 1/re^{-i\theta}$ (i.e. z_1 lies on the same ray as $\overline{z_0}$ but with larger magnitude). And so if z_0 were to lie in the lower half disc we would have that $z_0 + z_1$ has positive imaginary part. But as noted earlier, $w \in \mathbb{H}$ and so $-2w$ has negative imaginary part. Thus, we must have $z_0 \in \tilde{\Delta}$.

We have shown that given $w \in \mathbb{H}$ the equation $0 = z^2 + 2wz + 1$ has a unique solution in $\tilde{\Delta}$. In other words, we have shown that there exists a function $\mathbb{H} \to \tilde{\Delta}$ which inverts the given f. Hence f is a holomorphic bijection, thus f is a biholomorphism $\tilde{\Delta} \to \mathbb{H}$.

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